Hyperbolic Functions

Questions 1–111

1.30 — Question 1. Simplify each of these, writing your answers in exponential form:

a) $2\sinh x$
b) $8\cosh x$
c) $4\sinh x - 4\cosh x$
d) $5\cosh x - 10\sinh x$

1.30 — Question 2. Show that:

a) $2\sinh x + 4\cosh x \equiv e^{x}(3e^{-2x} + 1)$
b) $3\sinh x - 9\cosh x \equiv -3e^{-x}(e^{2x} + 2)$
c) $10\cosh x - 8\sinh x \equiv e^{-x}(e^{2x} + 9)$
d) $26\cosh x + 42\sinh x \equiv 2e^{x}(17e^{-2x} - 4)$

1.30 — Question 3. Show that $55\sinh x - 35\cosh x$ can be written in the form $ae^{-x}(be^{2x} - 9)$, where a and b are constants to be found.

1.30 — Question 4. By using the exponential forms of $\sinh x$ and $\cosh x$, find the value of each of these:

a) $\sinh 0$

b) $\cosh 0$

1.30 — Question 5. Write down the exact value of each of these in terms of exponentials, and then to 3 significant figures:

a) $\sinh 1$
b) $\cosh 1$
c) $\sinh(-1)$
d) $\cosh(-1)$

1.30 — Question 6. Show that $\sinh(-x) = -\sinh x$.

1.30 — Question 7. Show that $\cosh(-x) = \cosh x$.

1.30 — Question 8. By using the exponential forms of $\sinh x$ and $\cosh x$, find the value of each of these:

a) $\sinh(\ln 2)$
b) $\cosh(\ln 3)$
c) $\sinh(\ln(1\frac{1}{3}))$
d) $\cosh(\ln(1\frac{1}{4}))$

1.30 — Question 9. Show that

$$\sinh(\ln k) = \frac{k^2 - 1}{2k}$$

where k is a positive constant.

1.30 — Question 10. Show that

$$\cosh(2\ln k) = \frac{k^4 + 1}{2k^2}$$

where k is a positive constant.

1.30 — Question 11. By using the exponential forms of $\sinh x$ and $\cosh x$, show that

$$\tanh x \equiv \frac{(e^x+1)(e^x-1)}{e^{2x}+1}$$

1.30 — Question 12. By using the exponential forms of $\sinh x$ and $\cosh x$, show that

$$\frac{1}{2}\left(\frac{1+\tanh x}{1-\tanh x}\right) = x$$

1.30 — Question 13. Write down the exact value of each of these in terms of exponentials, and then to 3 significant figures:

a) $\tanh 0$
b) $\tanh 1$
c) $\tanh(-1)$

1.30 — Question 14. Show that $\tanh(-x) = -\tanh x$.

1.30 — Question 15. By using the exponential form of $\tanh x$, find the value of each of these:

a) $\tanh(\ln 3)$

b) $\tanh(-\ln 4)$

c) $\tanh(\ln \frac{1}{5})$

1.30 — Question 16. Prove, for all real values of x, that $\sinh x + \cosh x > 0$.

1.30 — Question 17. Prove, for all real values of x, that $\sinh x - \cosh x < 0$.

1.30 — Question 18. Use exponentials to solve the following equations, writing your answers in exact form:

a) $\sinh x = 0$

b) $\sinh x = 1$

c) $\sinh x = -10$

1.30 — Question 19. Use exponentials to solve the following equations, writing your answers in exact form:

a) $\cosh x = 1$

b) $\cosh x = 5$

c) $\cosh x = 23$

1.30 — Question 20. Use exponentials to solve the following equations, writing your answers in exact form:

a) $\tanh x = 0$

b) $\tanh x = \frac{1}{3}$

c) $\tanh x = -\frac{4}{5}$

1.30 — Question 21. Solve the following equations, writing your answers in exact form:

a) $\sinh x + \cosh x = 2$

b) $2\sinh x - \cosh x = 1$

c) $\sinh x - 2\cosh x = -2$

1.30 — Question 22. Solve the following equations, writing your answers in exact form:

a) $2\cosh x - 3\sinh x = 1$

b) $5\sinh x - 6\cosh x = -7$

c) $3\cosh x - 5\sinh x = 2$

d) $4\sinh x + 3\cosh x = 4$

1.30 — Question 23. Solve the following equations, writing your answers in exact form:

a) $2\sinh 2x = 2 - 7e^{2x}$

b) $3\sinh 2x = 2e^{2x} - 2$

1.30 — Question 24.

a) Fully expand $(y^2 - 6y + 1)(y^2 + y + 1)$.

b) Use exponentials to solve $\cosh 2x = 2 + 5\cosh x$, writing your answers in exact form.

1.30 — Question 25.

a) It is given that $f(y) = y^4 - 4y^3 + 4y - 1$.

(i) Show that $f(1) = 0$ and $f(-1) = 0$.

(ii) Fully factorise $f(y)$.

b) Use exponentials to solve $\sinh 2x = 4 \sinh x$, writing your answers in exact form.

1.30 — Question 26. Solve the following equations, writing your answers in exact form:

a) $\sinh^2 x + \sinh x - 6 = 0$

b) $8 \cosh^2 x - x - 26 \cosh x + 15 = 0$

1.30 — Question 27. Solve the following equations, writing your answers in exact form:

a) $\cosh^3 x - 5\cosh^2 x - 2\cosh x + 24 = 0$

b) $12\cosh^4 x - 44\cosh^2 x - 45 = 0$

1.30 — Question 28. Solve the following equations, writing your answers in exact form:

a) $15\tanh^2 x - 8\tanh x + 1 = 0$

b) $12\tanh^2 x - 17\tanh x + 6 = 0$

1.30 — Question 29.

a) It is given that $f(y) = y^3 - 7y^2 + 7y - 1$.

(i) Show that $f(1) = 0$.

(ii) Fully factorise $f(y)$.

b) Show that $x = 0$, $x = \ln(\sqrt{2} - 1)$ and $x = \ln(1 + \sqrt{2})$ are the real roots of the equation $3\tanh 2x = 4\tanh x$.

1.30 — Question 30. Solve the following equations, writing your answers in exact form:

a) $\frac{1}{\cosh x} + \frac{1}{\sinh x} = e^{3x}$

b) $\frac{1}{\cosh x} + \frac{1}{\sinh x} = \frac{4}{e^x}$

c) $\frac{1}{\cosh x} + \sinh x = \frac{2}{e^x}$

d) $4\cosh x - \frac{1}{\sinh x} = \frac{2}{e^x}$

1.30 — Question 31.

a) It is given that $f(y) = y^4 + 8y^3 + 2y^2 - 16y - 11$.

(i) Show that $f(-1) = 0$.

(ii) Given that $(y+1)^2$ is a factor of $f(y)$, fully factorise $f(y)$.

b) Use exponentials to solve

$$\frac{2\sinh x}{1+\cosh x} + \frac{1+\cosh x}{2\sinh x} = 3$$

writing your answers in exact form.

1.30 — Question 32. Show that

$$\cosh^3 x + \sinh 3x = \frac{1}{8}(e^{3x} + 3e^x + 3e^{-x} - e^{-3x})$$

where a is a positive integer.

1.30 — Question 33.

a) Show that

(i) $\sinh^3 x - 4\cosh^3 x = -(e^{-3x} + ae^x)$ where a is a positive integer.

(ii) $4\sinh^3 x + 4\cosh^3 x = e^{3x} + be^{-x}$ where b is a positive integer.

b) Hence show that

$$16\sinh^6 x - 16\cosh^6 x = -2(p\cosh 4x + q)$$

where p and q are positive integers.

1.30 — Question 34. Prove $(\sinh x + \cosh x)^3 = e^{3x}$.

1.30 — Question 35.

a) By using the exponential forms of $\cosh x$, prove that $4\cosh^3 x \equiv \cosh 3x + 3\cosh x$.

b) Hence show that the equation

$$4y^3 - 3y = \frac{5}{4}$$

has a real solution of the form $p^{1/4} + q^{1/4}$, where p is an integer and q and r are rational numbers.

c) Use this result to find the exact values of p, q, r.

Hyperbolic Identities

1.30 — Question 36. Simplify the following expressions as far as possible, writing your answers in terms of $\sinh x$ and $\cosh x$ only:

a) $\cosh x \tanh x$

b) $\frac{\tanh x}{\sinh x}$

c) $\frac{\tanh x}{\cosh x}$

1.30 — Question 37. Simplify the following expressions as far as possible, writing your answers in terms of $\sinh x$ and $\cosh x$ only:

a) $\sinh x \cosh x \tanh x$

b) $\frac{\sinh^2 x \cosh x \tanh^2 x}{\sinh x}$

c) $\frac{\sinh x}{\cosh^2 x \tanh x}$

d) $\frac{\tanh^2 x}{\sinh^2 x \cosh x}$

1.30 — Question 38. By using the exponential forms of $\sinh x$ and $\cosh x$, prove that

$$\cosh^2 x - \sinh^2 x = 1$$

1.30 — Question 39. By using the exponential forms of $\sinh x$ and $\cosh x$, prove that

$$\sinh 2x = 2\sinh x \cosh x$$

1.30 — Question 40. By using the exponential forms of $\sinh x$ and $\cosh x$, prove that:

a) $\cosh 2x = \cosh^2 x + \sinh^2 x$

b) $\cosh 2x = 2\cosh^2 x - 1$

c) $\cosh 2x = 1 + 2\sinh^2 x$

1.30 — Question 41. By using the exponential forms of $\sinh x$ and $\cosh x$, prove that

$$\tanh 2x = \frac{2\tanh x}{1+\tanh^2 x}$$

1.30 — Question 42. Write each of these in terms of $\sinh x$ only:

a) $5\cosh^2 x + 3\sinh^2 x$

b) $2\sinh^2 x - 6\cosh^2 x$

1.30 — Question 43. Write each of these in terms of $\cosh x$ only:

a) $9\cosh^2 x + 2\sinh^2 x$

b) $5\sinh^2 x - 8\cosh^2 x$

1.30 — Question 44. Given that $\cosh 2x = 2\cosh^2 x - 1$ and $\cosh 2x = 1 + 2\sinh^2 x$, write each of these in terms of $\cosh 2x$ only:

a) $2\cosh^2 x + 5\sinh^2 x$

b) $8\sinh^2 x - 3\cosh^2 x$

1.30 — Question 45. Use hyperbolic identities to solve the following equations, writing your answers in exact form:

a) $\cosh^2 x + \sinh^2 x = 3$
b) $2\cosh^2 x + 3\sinh x - 4 = 0$
c) $5\sinh x = 2\cosh^2 x$
d) $3\sinh^2 x = 2\cosh x + 5$

1.30 — Question 46. Use hyperbolic identities to solve the following equations, writing your answers in exact form:

a) $\cosh 2x = 3 + 7\cosh x$
b) $5\cosh 2x + 8 = \sinh x - 7$
c) $8\cosh 2x = \cosh x + 9$
d) $\sinh x = 5 - 2\cosh 2x$

1.30 — Question 47. Use hyperbolic identities to solve the following equations, writing your answers in exact form:

a) $\sinh 2x = 20 \sinh x$

b) $\frac{2}{3} \sinh 2x = \cosh x$

1.30 — Question 48. Dylan is trying to solve the equation $3 \sinh x + \cosh^2 x = 5$. His working is as follows:

$$3 \sinh x + 1 - \sinh^2 x = 5$$

$$\Rightarrow 0 = \sinh^2 x - 3 \sinh x + 4$$

$$\Rightarrow 0 = (\sinh x - \frac{3}{2})^2 - \frac{9}{4} - 4$$

$$\Rightarrow 0 = (\sinh x - \frac{3}{2})^2 - \frac{25}{4}$$

$$\Rightarrow \sinh x = \frac{3}{2} \pm \frac{5}{2} = 4 \text{ or } -1$$

$$\Rightarrow x = \sinh^{-1} 4 \text{ or } \sinh^{-1}(-1)$$

a) Identify the two errors Dylan has made in his working.

b) Write out a correct method and find the exact values of x that solve the equation.

1.30 — Question 49.

a) Prove $8\cosh^4 x \equiv \cosh 4x + 4\cosh 2x + 3$.

b) Solve the equation $2\cosh 4x - 23\cosh 2x - 33 = 0$.

1.30 — Question 50. Solve the following equations, writing your answers in exact form:

a) $\sinh x(\sinh x + \cosh x) = 4$

b) $\cosh x(\sinh x + \cosh x) = 13$

c) $\sinh x(5\sinh x - 4\cosh x) = 3$

1.30 — Question 51. Use hyperbolic identities to solve the following equations, writing your answers in exact form:

a) $3\tanh 2x - 4\tanh x = 0$

b) $10\tanh 2x = 9 + \tanh x$

1.30 — Question 52. By using the exponential forms of $\sinh x$ and $\cosh x$, prove that

$$\sinh x \cosh y + \cosh x \sinh y \equiv \sinh(x + y)$$

1.30 — Question 53. By using the exponential forms of $\sinh x$ and $\cosh x$, prove that

$$\cosh x \cosh y - \sinh x \sinh y \equiv \cosh(x - y)$$

1.30 — Question 54. It is given that $\sinh(x + \ln 5) = \cosh x$.

a) Show that $\tanh x = -\frac{7}{13}$.

b) Hence show that $x = \frac{1}{2}(\ln 3 - \ln 2 - \ln 5)$.

1.30 — Question 55. Use the exponential form of $\sinh x$ to prove

$$\sinh 3x = 4\sinh^3 x + 3\sinh x$$

1.30 — Question 56. Use the exponential form of $\sinh x$ to prove

$$\cosh 3x = 4\cosh^3 x - 3\cosh x$$

1.30 — Question 57. Use the identities proved in the previous two questions to solve the following equations:

a) $\sinh 3x = 15 \sinh x$

b) $\cosh 3x = 5 \cosh x$

1.30 — Question 58. Given that $\sinh x = 3$, where $x > 0$, find the exact value of:

a) $\cosh x$

b) $\tanh x$

1.30 — Question 59. Given that $\cosh x = 5$, where $x > 0$, find the exact value of:

a) $\sinh x$

b) $\tanh x$

1.30 — Question 60. Given that $\tanh x = \frac{1}{3}$, where $x > 0$, find the exact value of:

a) $\sinh x$

b) $\cosh x$

1.30 — Question 61. Find the value(s) of x that satisfy both of the following equations:

$$2\cosh x + \sinh x = 2$$

$$4\cosh^2 x - 11\sinh x = 4$$

1.30 — Question 62. Find the value(s) of x that satisfy both of the following equations:

$$2\cosh^2 x - \sinh x = 8$$

$$2\cosh^2 x - 3\sinh x = 4$$

1.30 — Question 63. Prove the following identities:

a) $(\sinh x + \cosh x)^2 \equiv \sinh 2x + \cosh 2x$

b) $(\sinh 2x - \cosh 2x)^2 \equiv \cosh 4x - \sinh 4x$

c) $1 - \tanh^2 x \equiv \frac{1}{\cosh^2 x}$

d) $\frac{1}{\tanh^2 x} - 1 \equiv \frac{1}{\sinh^2 x}$

1.30 — Question 64. Prove

$$\sinh^3 x = \frac{1}{4}\sinh 3x - \frac{3}{4}\sinh x$$

1.30 — Question 65. Prove

$$\cosh 3x - 3\cosh x = 2\cosh x(\cosh 2x - 2)$$

1.30 — Question 66. Prove

$$\cosh 2x - 2\sinh^2 x \equiv 1$$

1.30 — Question 67. Prove

$$3\sinh x - 7\sinh(\frac{x}{2}) = \sinh(\frac{x}{2})(6\cosh(\frac{x}{2}) - 7)$$

1.30 — Question 68. Prove

$$\sinh x = 4\sinh(\frac{x}{4})\cosh(\frac{x}{4})\cosh(\frac{x}{2})$$

1.30 — Question 69. Prove

$$\cosh^2(\frac{x}{3}) - 4\sinh^2(\frac{x}{6})\cosh^2(\frac{x}{6}) \equiv 1$$

1.30 — Question 70. Prove

$$4\cosh(\frac{x}{2}) = 4\cosh^4(\frac{x}{4}) - \sinh^2(\frac{x}{2})\tanh^2(\frac{x}{4})$$

1.31 — Question 1. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

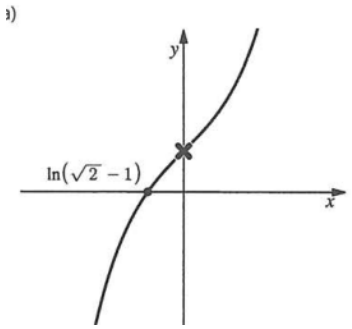
a) $y = \sinh x$
b) $y = \cosh x$
c) $y = \tanh x$

1.31 — Question 2. Sketch the following pairs of graphs on the same axes. Label each graph clearly and show where they intersect the coordinate axes:

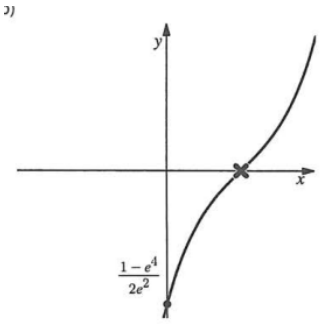
a) $y = \sinh x$ and $y = 2\sinh x$
b) $y = \sinh x$ and $y = \sinh(x + 3)$
c) $y = \sinh x$ and $y = -\sinh x$
d) $y = \cosh x$ and $y = \cosh x + 1$
e) $y = \cosh x$ and $y = \cosh 2x$
f) $y = \cosh x$ and $y = -\cosh x$

1.31 — Question 3. By considering the point of inflection (identified on the graphs by a ×), find the equation of each of the following graphs in the form $y = \sinh(x - a)$ or $y = \sinh x + b$, where a and b are integers:

a)

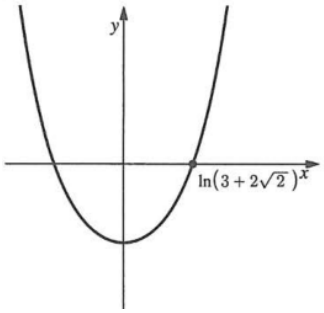


b)

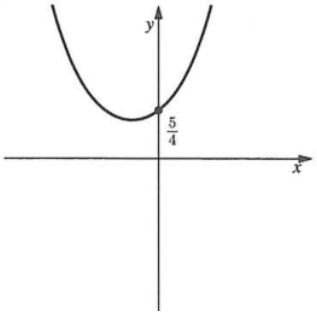


1.31 — Question 4. Find the equation of each of the following graphs in the form $y = \cosh(x-a)$ or $y = \cosh x+b$, where a and b are integers:

a)



b)



1.31 — Question 5. State the equations of the asymptotes of the following graphs:

a) $y = \tanh(x - 2) + 5$

b) $y = \tanh(x + \frac{5}{2}) - \frac{3}{2}$

1.31 — Question 6. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \tanh x + 1$

b) $y = \tanh(x - 1)$

c) $y = \tanh x - 2$

d) $y = -\tanh x$

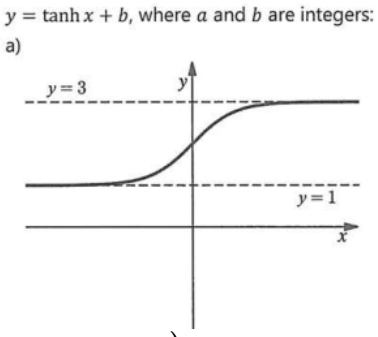
1.31 — Question 7. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = 3\tanh x$

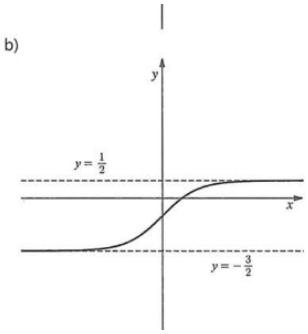
b) $y = \frac{1}{2}\tanh x$

1.31 — Question 8. Find the equation of each of the following graphs in the form $y = \tanh(x-a)$ or $y = \tanh x + b$, where a and b are integers:

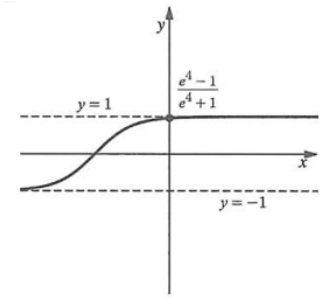
a)



b)



c)



1.31 — Question 9. It is given that $k > 1$. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

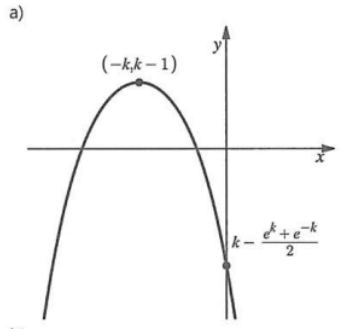
a) $y = \sinh(x - k) + k$
b) $y = \cosh(x + k) - k$
c) $y = \tanh(x - k) - k$

1.31 — Question 10. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

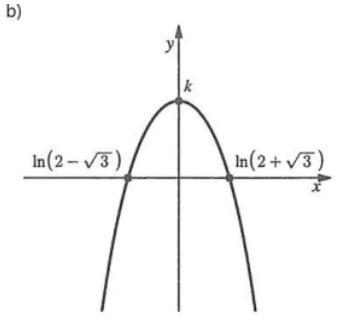
a) $y = 1 - \sinh x$
b) $y = -\frac{1}{2}\cosh x$
c) $y = 2\tanh x + 1$
d) $y = 2\sinh x - 1$
e) $y = 3\cosh x - 5$
f) $y = 2 - \frac{1}{2}\tanh x$

1.31 — Question 11. Both of the following hyperbolic curves are transformations of $y = \cosh x$. No horizontal stretches have been performed. For constant k where $k > 1$, find a suitable equation for each:

a)



b)



1.31 — Question 12. It is given that $1 < k < 2$. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \sinh(kx - 2k)$
b) $y = 3k - \cosh(x + k)$
c) $y = 4k - k\tanh(x - k)$

1.32 — Question 1. Given $y = \sinh x$, show that

$$\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$$

1.32 — Question 2. Given $y = \cosh x$, show that

$$\operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1})$$

stating the values of x for which $\operatorname{arcosh} x$ exists.

1.32 — Question 3. Given $y = \tanh x$, show that

$$\operatorname{artanh} x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

stating the values of x for which $\operatorname{artanh} x$ exists.

1.32 — Question 4. Write down the exact values of the following:

a) $\operatorname{arsinh} 1$
b) $\cosh^{-1} 2$
c) $\operatorname{artanh} \frac{1}{3}$
d) $\sinh^{-1}(-4)$
e) $\operatorname{arcosh} 9$
f) $\tanh^{-1}(-\frac{1}{2})$

1.32 — Question 5. Solve the following equations, writing your answers in exact form:

a) $\sinh x = 0$

b) $\sinh x = 3$

c) $\sinh x = -6$

1.32 — Question 6. Solve the following equations, writing your answers in exact form:

a) $\cosh x = 1$

b) $\cosh x = 6$

c) $\cosh x = \frac{3}{2}$

1.32 — Question 7. Solve the following equations, writing your answers in exact form:

a) $\tanh x = 0$
b) $\tanh x = \frac{1}{4}$
c) $\tanh x = -\frac{5}{6}$

1.32 — Question 8. Solve the following equations, writing your answers in exact form:

a) $\sinh 2x = 5$
b) $\cosh 3x = 8$
c) $5\tanh(3x - 1) = 2$
d) $3\sinh(2 - \frac{1}{3}x) = 1$

1.32 — Question 9.

a) Sketch and label the graphs of $y = \sinh x$ and $y = \cosh x$ on the same axes.

b) Prove that the graphs of $y = \sinh x$ and $y = \cosh x$ do not intersect.

1.32 — Question 10.

a) Sketch and label the graphs of $y = \cosh x$ and $y = \tanh x$.

b) Given that $\cosh^2 x \equiv 1 + \sinh^2 x$, prove that the graphs of $y = \cosh x$ and $y = \tanh x$ do not intersect.

1.32 — Question 11. Solve the following equations, writing your answers in exact form:

a) $\sinh^2 x + \sinh x - 12 = 0$

b) $2\cosh^2 x - \cosh x - 21 = 0$

c) $12\tanh^2 x + \tanh x - 1 = 0$

1.32 — Question 12. Solve the equation $3\cosh^2 x + 10\cosh x - 25 = 0$ writing your answers in the form $\ln k$, where k is a rational number.

1.32 — Question 13. Solve the equation $3\cosh^2 x - 17\cosh x - 56 = 0$ writing your answers in the form $\ln(a+b\sqrt{7})$, where a and b are integers.

1.32 — Question 14. Solve the following equations:

a) $\operatorname{arsinh} x = \ln 2$
b) $\operatorname{arcosh} x = \ln 3$
c) $\operatorname{artanh} x = \ln 4$
d) $\operatorname{arsinh} 2x = 2\ln 3$
e) $\operatorname{arcosh} 3x = 3\ln 2$
f) $\operatorname{artanh} \frac{1}{2}x = 2\ln \frac{1}{2}$

1.32 — Question 15. Solve the following equations:

a) $x = \sinh(\ln(3 - x))$

b) $\cosh(\ln(2x - 1)) = x + 2$

c) $x - 1 = \tanh(\ln\sqrt{x + 1})$

1.32 — Question 16. Solve the following equations:

a) $\operatorname{artanh} x - \ln(1 + x) = \ln\frac{6}{5}$

b) $\ln 2 + \ln(1 + x) - \operatorname{artanh} x = \ln\frac{2}{9}$

c) $\operatorname{arsinh} x - \ln(1 + x) = \ln 2$

d) $\ln 3 + \ln(x - 1) - \operatorname{arcosh} x = 0$

1.32 — Question 17. Solve the following equations:

a) $\operatorname{arsinh} x + \ln 2 = \operatorname{arcosh} x + \ln 3$

b) $\operatorname{arsinh} x - 2\operatorname{arcosh} x = 0$

1.33 — Question 1. Sketch the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = \operatorname{arsinh} x$

b) $y = \operatorname{arcosh} x$

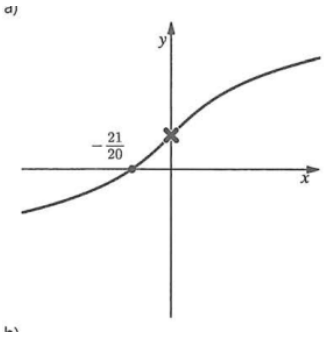
c) $y = \operatorname{artanh} x$

1.33 — Question 2. Sketch the following pairs of graphs on the same axes. Label each graph clearly and show where they intersect the coordinate axes:

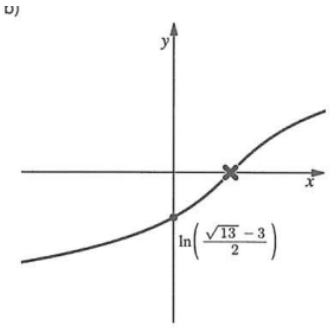
a) $y = \operatorname{arsinh} x$ and $y = 3\operatorname{arsinh} x$
b) $y = \operatorname{arsinh} x$ and $y = \operatorname{arsinh}(x - 2)$
c) $y = \operatorname{arsinh} x$ and $y = -\operatorname{arsinh} x$
d) $y = \operatorname{arcosh} x$ and $y = \operatorname{arcosh} \frac{1}{2}x$
e) $y = \operatorname{arcosh} x$ and $y = \operatorname{arcosh}(x + 3)$
f) $y = \operatorname{arcosh} x$ and $y = \operatorname{arcosh}(-x)$

1.33 — Question 3. By considering the point of inflection (identified on the graphs by a $\times$), find the equation of each of the following graphs in the form $y = \operatorname{arsinh}(x - a)$ or $y = \operatorname{arsinh} x + b$, where a and b are integers:

a)

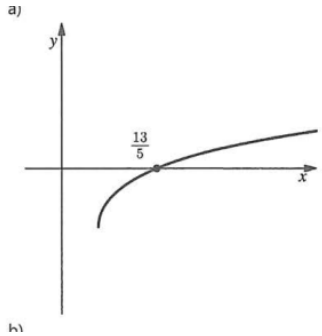


b)

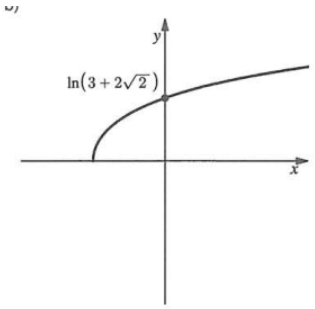


1.33 — Question 4. Find the equation of each of the following graphs in the form $y = \text{arcosh}(x - a)$ or $y = \text{arcosh}\, x + b$, where a and b are integers:

a)



b)



1.33 — Question 5. State the equations of the asymptotes of the following graphs:

a) $y = \text{artanh}(x + 5) - 3$

b) $y = \text{artanh}(x - \frac{1}{3}) + 2$

1.33 — Question 6. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

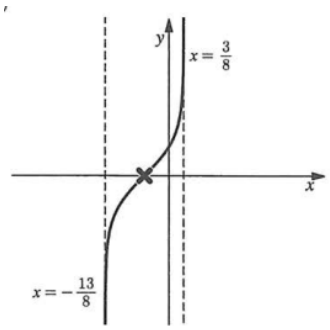
a) $y = \operatorname{artanh} x - 1$
b) $y = \operatorname{artanh}(x + 3)$
c) $y = \operatorname{artanh}(x - \frac{1}{2})$
d) $y = -\operatorname{artanh} x$

1.33 — Question 7. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

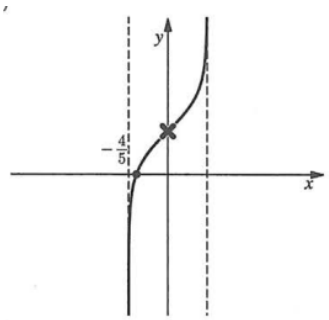
a) $y = \operatorname{artanh} 3x$
b) $y = \operatorname{artanh} \frac{1}{4}x$

1.33 — Question 8. In each of the following graphs, the point of inflection is identified by a $\times$. Find the equation of each in the form $y = \operatorname{artanh}(x - a)$ or $y = \operatorname{artanh} x + b$, where a and b are integers:

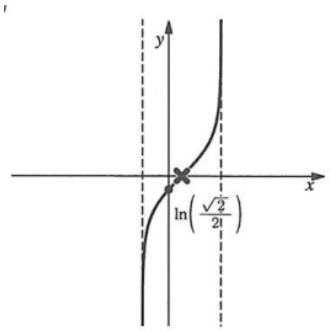
a)



b)



c)



1.33 — Question 9. Sketch the following graphs, given $k > 1$, identifying where they intersect the coordinate axes and the equations of any asymptotes:

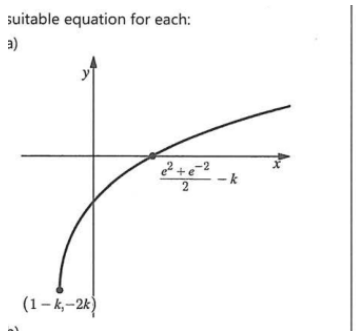
a) $y = \operatorname{arsinh}(x - k) + k$
b) $y = \operatorname{arcosh}(x + k) - k$
c) $y = \operatorname{artanh}(x - k) - k$

1.33 — Question 10. Sketch each of the following graphs, identifying where they intersect the coordinate axes and the equations of any asymptotes:

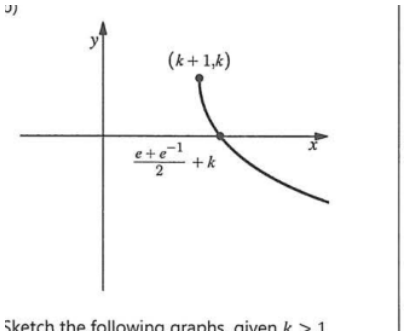
a) $y = 2 - \operatorname{arsinh} x$
b) $y = -\operatorname{arcosh} 2x$
c) $y = \operatorname{artanh}(\frac{1}{2}x + 1)$
d) $y = \operatorname{arsinh}(1 - x)$
e) $y = -\operatorname{arcosh}(-x)$
f) $y = -\operatorname{artanh} 2x$

1.33 — Question 11. Both of the following inverse hyperbolic curves are transformations of $y = \operatorname{arcosh} x$. No horizontal stretches have been performed. Given k is a constant where $1 < k < 3$, find a suitable equation for each:

a)



b)



1.33 — Question 12. Sketch the following graphs, given $k > 1$, identifying where they intersect the coordinate axes and the equations of any asymptotes:

a) $y = k - \operatorname{arsinh}(x + k)$
b) $y = \operatorname{arcosh}(x - 3k)$
c) $y = k - \operatorname{artanh}(kx + k)$